

UTM TENSILE REPORT — UTM_Test_20260824_113342

DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A_0) · anchor self-calibration

48.3 MPa

UTS

39.3 MPa

Yield σ_y

3.45 GPa

Modulus E

5.3 %

Failure ϵ_f

2184 kJ/m³

Toughness

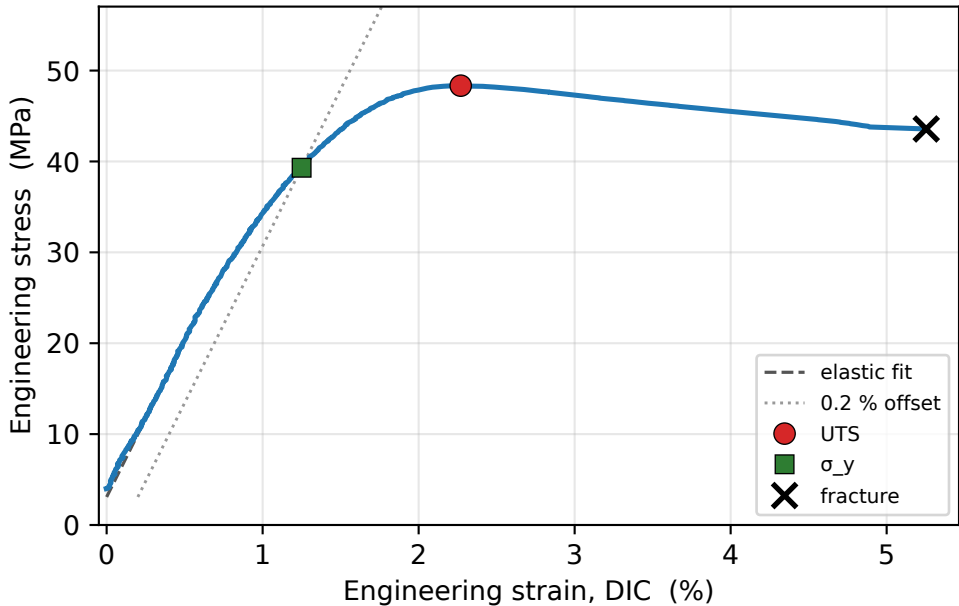
318 N

Preload anchor

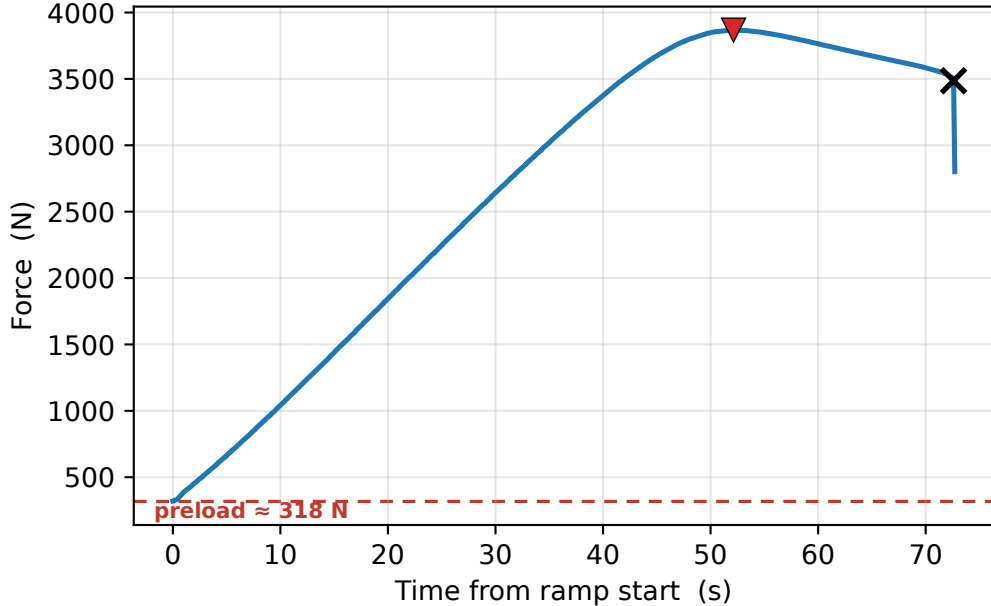
TEST SETTINGS

Specimen	UTM_Test_20260824_113342
DIC mode	White
Preload target	300 N
Test speed	0.100 mm/s
Cross-section	80 mm ²
Gauge length L ₀	80 mm
Infill (label)	99 %
Force calib	sc -0.0065 / off 341.926
Comment	—
Test date	2026-08-24 11:32:07
Duration	104.5 s

Stress-strain



Load vs time

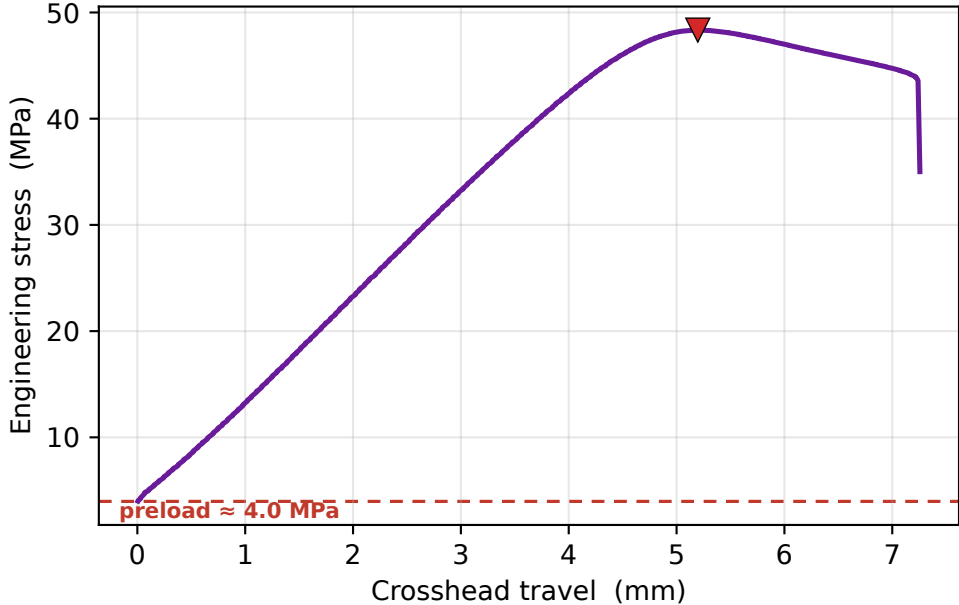


VALIDATION vs REFERENCES

prop	meas	Chacón	E-PLA k
UTS	48.3	32-60 PASS	×1.20
σ_y	39.3	30-50 PASS	×1.48
E	3.5	3-5.5 PASS	×0.83
ϵ_f	5.3 %	(range 3-7 %)	×1.52

✓ strength inside journal range ($k \approx 1$)

Stress vs crosshead



DIC strain vs crosshead — the compliance gap

